



TSMC Macrocells Data Sheet

PG13A1G3

1GHz PLL-Based Clock Generator

Date : 11/21 ,2003

Version	Date	Hspice Netlist Update	GDSII Update	Design Rule Version	Verilog Model Update	Synopsys Model Update
100a	6/14/2001	Original	Original	1.0	N/A	N/A
110a	9/07/2001	Yes	Yes	1.1	Yes	Yes
110b	10/12/2001	Yes	Yes	1.1	No	No
110c	11/19/2001	Yes	Yes	1.1	No	No
120b	3/21/2002	No	No	1.2	No	No
150a	11/15/2002	Yes	Yes	1.5	No	No
200a	11/21/2002	No	Yes	2.0	No	No

Release Note

- 06/14/01' : Original prototype.
- 07/23/01' : DIV pin is removed, and two modifications are made to improve reliability ...
 - The expression for NF is changed to be

$$NF = (3-NO)*(32*F5+16*F4+8*F3+4*F2+2*F1+F0+2)$$
 - Constraints on divider setting are imposed

$$NF \geq 6*(3-NO)$$
- 08/31/01' : FOUT state is changed to be uncertain (0 or 1) when PD=1.
- 09/07/01' : Update Long Term Jitter spec.
- 10/12/01' : Design kit update.
- 11/19/01' : To comply with design rule V1.2, and improve ESD robustness.
- 03/21/02' : Add constraint on F_{IN} Update rise time, Tsr, and pin capacitances.
- 11/15/02' : To comply with DRC 1.5. Improve power down quality and ESD robustness.
- 11.21/03' : To comply with DRC1.5.

Technology

PG13A1G3 is a 1P4M layout structure based on TSMC 0.13um 1P4M~1P8M 1.2V/3.3V generic logic process. In the layout of this cell, the vias between metal-2 and metal-3 are of normal size. To fit the top-via specification of design rule which enlarge the vias connecting to the top metal, we suggest our customer using PG13A1G3 in the chip which contains at least 4 layers of metal.

Features

- Dual power supplies 1.2V/3.3V required.
- Fully integrated PLL-based clock generator including loop filter.
- VCO operating frequency is around 500 ~ 1000 MHz.
- PFD comparison frequency range: 10 ~ 50MHz. Input frequency range is determined by comparison frequency and input divider setting.
- Low jitters.
- Macro cell Size: 350.340um x 376.435um.

General Description

PG13A1G3 is developed as a PLL-based macro cell for clock generator. The output frequency is adjustable and the range is up to 1GHz.

Pin Information

Pin address

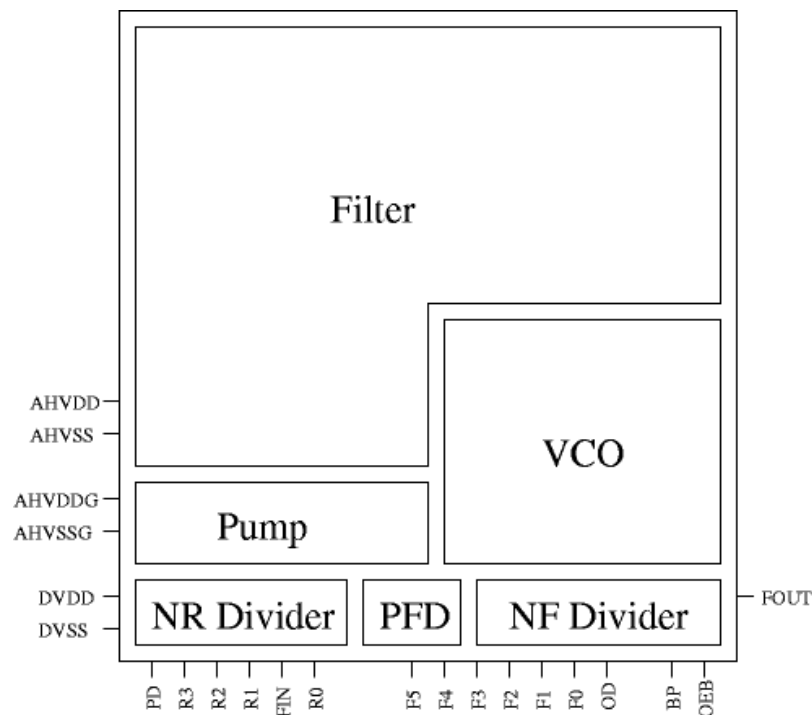


Fig. 1 PG13A1G3 Pin Address

Pin Definition

PIN NAME	PIN TYPE	DESCRIPTION	INPUT CAPACITANCE (fF) & OUTPUT SLEW RATE(ns)
AHVDD	POWER	Dedicated Analog Power 3.3V	[1,2]
AHVSS	POWER	Dedicated Analog Ground	[1,2]
AHVDDG	POWER	Dedicated Analog Power 3.3V	[1,2,3]
AHVSSG	POWER	Dedicated Analog Ground	[1,2,3]
DVDD	POWER	Dedicated Digital Power 1.2V	[1,2]
DVSS	POWER	Dedicated Digital Ground	[1,2]
FIN	INPUT	Reference clock input	50.40
FOUT	OUTPUT	Adjustable clock output	$1.85 \cdot CL + 0.050$ CL in pF][8]
OEB	INPUT	FOUT enable pin	21.12
BP	INPUT	Bypass the PLL	23.49
PD	INPUT	Power down mode	127.59
OD	INPUT	Output divider control pin	45.48
R3	INPUT	Input 4-bit divider control pin (MSB)	32.71
R2	INPUT	Input 4-bit divider control pin	31.49
R1	INPUT	Input 4-bit divider control pin	29.65
R0	INPUT	Input 4-bit divider control pin (LSB)	26.67
F5	INPUT	Feedback 6-bit divider control pin (MSB)	104.02
F4	INPUT	Feedback 6-bit divider control pin	103.48
F3	INPUT	Feedback 6-bit divider control pin	100.48
F2	INPUT	Feedback 6-bit divider control pin	96.40
F1	INPUT	Feedback 6-bit divider control pin	35.98
F0	INPUT	Feedback 6-bit divider control pin (LSB)	43.45

Netlist Order

MODULE PG13A1G3(FOUT, AHVDD, AHVSS, AHVDDG, AHVSSG, DVDD,
DVSS, BP, F0, F1, F2, F3, F4, F5, FIN, OD, OEB, PD, R0, R1, R2, R3);
[1,2,3,4]

INPUT FIN, OEB, BP, PD, OD, F0, F1, F2, F3, F4, F5, R0, R1, R2, R3;

OUTPUT FOUT;

INOUT DVDD, DVSS, AHVDD, AHVSS, AHVDDG, AHVSSG [1,2,3];

Note [1]: All Power pins are treated as "signals". This is to distinguish the CORE VDD & VSS from our power pins, which need special treatment in P&R and LVS tools.

Note [2]: The voltage values are different for different power-ground pairs. DVDD, DVSS pair uses 1.2V while AHVDD, AHVSS pair uses 3.3V.

Note [3]: The power pin AHVDDG is also 3.3V but it is very noisy, so please do NOT share the AHVDDG power with the other 3.3V power AHVDD. AHVSSG also doesn't share with AHVSS.

DC Characteristics

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT	CONDITION
Operating voltage range	DVDD	1.08	1.2	1.32	V	
Operating voltage range	AHVDD AHVDDG	2.97	3.3	3.63	V	
Input low voltage	VIL	DVSS	--	0.2DVDD	V	DVDD = 1.2V
Input high voltage	VIH	0.8DVDD	--	DVDD	V	DVDD = 1.2V

AC Characteristics

PARAMETER	SYM	MIN	TYP	MAX	UNIT	CONDITION
Comparison Frequency	F_{REF}	10	--	50	MHz	$F_{REF} = F_{IN}/NR$
Input clock Frequency	F_{IN} [4,5]			200	MHz	
Output clock Frequency	F_{OUT}	250	--	1000	MHz	$F_{OUT} = F_{VCO}/NO$ where NO is output divider value[5]
VCO Frequency	F_{VCO}	500	--	1000	MHz	
Period Jitter [6]	--	-50	--	+50	ps	Clean Power
Cycle-to-Cycle Jitter [6]	--	-80	--	+80	ps	Clean Power
Long Term Jitter [6]	--	-200	± 100	+200	ps	Clean Power Cycle=2048[7]
Static Phase Error Between F_{IN} and F_{OUT}	-	-1	--	+1	ns	At rising edge
Duty cycle	--	40	--	60	%	Without divide-by-2 (OD=0)
	--	45	--	55	%	After divide-by-2 (OD=1)

Note [4]: We suggest our customer to put buffers in front of F_{IN} pin to shape the input clock. Rise and fall time of the input clock being less than 1 ns is preferred.

Note [5]: Please refer to the **Basic Structure & Operation** section to see detail frequency relationship and divider setting.

Note [6]: Please refer to **Appendix A** to see the different jitter definitions. And given a jitter definition for measurement, the value is also different for different NF and NR divider's setting. Fin input jitter, divider setting, and power noise are the factors that affect the performance of PLL.

Note [7]: Long Term jitter's performance is strongly dependent on divider setting (loop bandwidth) and over how many cycles it should be measured.

Other Specifications

Parameter	Symbol	Min	Typical	Max	Unit	Condition
Silicon Area	--	0.13188			mm ²	350.340×376.435 um ²
Power Dissipation	--	--	3.8	12	mW	FOUT = 1GHz, NR=2, NF=40, 1.2V/3.3V, 25°C
Power Dissipation (Power down mode)	--	--	5	--	uW	PD=1, No FIN, 1.2V/3.3V, 25°C
Pull_in Time + LockingTime	Tready	--	--	0.5	ms	
FOUT Output Slew Rate [8]	Tsr	1.25*CL +0.034	1.85*CL +0.050	2.78*CL +0.070	ns	Simulation (CL in pf)

Note [8] :We define output slew rate as the output clock rising time delay from 0.1*DVDD to 0.9*DVDD (CL : Output Load in pF). The output loading condition should be carefully considered according to the clock output driving capability.

Basic Structure & Operation

Figure 2 shows the block diagram for PG13A1G3.

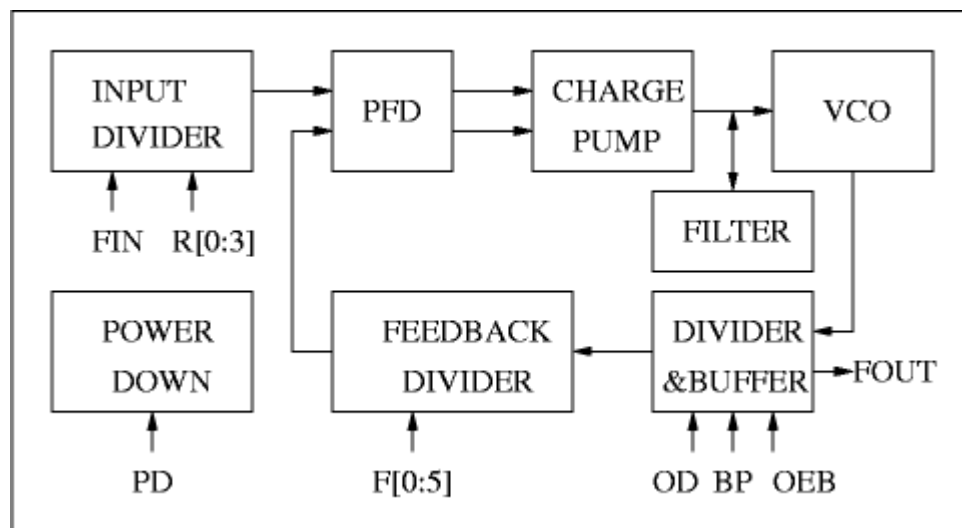


Fig. 2 PG13A1G3 Block Diagram

There are 3 operation modes in this macrocell: **Power Down Mode**, **Bypass Mode** and **Normal mode** (see Table 1). **Power Down Mode** forces FOUT to Low and make PLL in low power consuming state. **Bypass Mode** provides FOUT with the same frequency as FIN. **Normal Mode** synthesizes FOUT by programming suitable divider values. It needs a **Tready time** (Pull_in Time + Locking Time) for PLL to re-lock the FIN clock when PLL wakes up from Power Down Mode operation to Normal Mode. In general, it should be reserved a **Tready** time for re-locking when PLL is changed to Normal Mode from Power Down or Bypass Mode, or when any divider setting is changed.

Operation Mode Setting			FOUT	Description
PD	BP	OEB		
1	--	--	Uncertain, 0 or 1	Power Down Mode
0	1	0	FIN	Bypass Mode
0	0	0	--	Normal Mode
0	0	1	0	Output Disabled

Table 1. Operation Modes

Note that the logic level of **FOUT** is **uncertain** when **PD=1**.

In **Normal Mode** operation, it is necessary to set suitable divider values to make sure PLL functional:

- PLL Divider Value Setting

There are 3 divider values (NR, NF, NO) to set the PLL output clock frequency F_{OUT} :

- o Input Divider Value **NR**

$$NR = 8 \cdot R3 + 4 \cdot R2 + 2 \cdot R1 + R0 + 1$$

- o Feedback Divider Value **NF**

$$NF = (3 - NO) \cdot (32 \cdot F5 + 16 \cdot F4 + 8 \cdot F3 + 4 \cdot F2 + 2 \cdot F1 + F0 + 2)$$

- o Output Divider Value **NO**

$$NO = 1 + OD$$

When $OD=1$, F_{OUT} would have around 50% duty cycle.

- PLL Output Clock Frequency Setting

$$F_{OUT} = F_{IN} \cdot NF / NR$$

Meanwhile, the following **constraints** must be followed :

- o $10\text{MHz} \leq F_{REF} \leq 50\text{MHz}$

- o $500\text{MHz} \leq F_{VCO} \leq 1000\text{MHz}$

where

PLL's comparison frequency $F_{REF} = F_{IN}/NR$.

PLL's VCO frequency $F_{VCO} = F_{IN} \cdot NF \cdot NO / NR$.

- o $NF \geq 6 \cdot (3 - NO)$

- o Output clock F_{OUT} 's rise(fall) time should be less than 0.1ns.

For example, if we want to get a 300MHz output frequency clock (when using the 30MHz crystal oscillator as input),

NR=3, NF=30, NO=2, $F_{VCO}=600\text{MHz}$, $F_{OUT}=300\text{MHz}$:

F5:F0 = 30 -----> 0 1 1 1 0 0 (F5~F0)

R3:R0 = 3 -----> 0 0 1 0 (R3~R0)

OD = 1

The following examples (**Table 2**) show how to program the PLL's control pin and divider values in **Normal Mode**.

F_{IN} (MHz)	F_{OUT}	F_{VCO}	NF/NR	OD	R[3:0]	F[5:0]	Note
--	F_{VCO}	--	--	0	--	--	Normal Mode PD=0 BP=0 OEB=0
--	$F_{VCO}/2$	--	--	1	--	--	
50	250	500	25/5	1	0100	010111	
10	500	1000	50/1	1	0000	110000	
150	1000	1000	40/6	0	0101	010010	

Table 2. Normal Mode Examples

The PLL gives good performance based on proper divider settings. However, whether the divider setting is proper or not is determined by application. VCO stability and loop bandwidth play important roles on the quality of PLL's output clock. Of course one should give VCO (and whole PLL) clean power and pay more place-and-route consideration to avoid noise pollution. On the other hand, loop bandwidth selection (divider setting selection) may not be so straightforward. Narrower loop bandwidth (larger NF divider value, and usually with lower comparison frequency) makes PLL less sensitive to input clock jitter, leaves the PLL's performance simply determined by VCO. It usually gives good short-term jitter performance if the VCO is in a quiet circumstance. But in some applications the long-term jitter performance is of main concern. It is known that increasing the loop bandwidth (set lower NF value and increase the comparison frequency) could give good long-term jitter result. So there are different "proper settings" in different applications and situation (power, noise, and input clock quality). Please find the proper divider setting value according to application, and implement condition.

PD	FOUT
0	PLL normal operation
1	Uncertain, 0 or 1 [9]

Table 3. Power Down Mode Table

BP	FOUT
0	PLL normal operation
1	FIN

Table 4. Bypass Mode Table

Note [9]: While PLL wakes up from power down mode operation to normal operation, it needs **Tready time** (Pull_in Time + Locking Time) for PLL to re-lock the FIN clock.

OEB	FOUT
0	FOUT(= FVCO/NO)
1	0

Table 5. Output Control Pin Table

Analog Power Pads Assignment for PG13A1G3

- PG13A1G3 requires three power pads for (**DVDD, AHVDD, AHVDDG**), and three ground pads for (**DVSS, AHVSS, AHVSSG**).
- All power (**DVDD, AHVDD, AHVDDG**), and ground (**DVSS, AHVSS, AHVSSG**) should be considered as **analog power**. To Place-and-Route the PLL easily, all them are defined as signals rather than global powers. We strongly recommend our customer giving them dedicated power pads rather than sharing them with digital power (ground) of core logic. It could cause function failure at some logic patterns that have large number of switching if any power (ground) of PLL is shared by core logic.
- **DVDD is supplied with 1.2V power while AHVDD and AHVDDG are supplied 3.3V power.** Since the **AHVDDG** power is very noisy, please **DO NOT** share **AHVDD** the same I/O pad with **AHVDDG**. If the package pin counts are limited, double bonding the **AHVDD (AHVSS)** and **AHVDDG (AHVSSG)** to the same lead is a possible solution.
- Dedicated leads or balls on the package are recommended for the PG13A1G3 power/ground pads. **Do not** share PLL's power/ground with any other power/ground rings on the package.
- Please use the power cut cells (e.g, **PRDIODE**) to separate the analog power & ground plane from digital power & ground plane, you can use (**PVDD1PC, PVDD1P**) pad for (**DVDD, AHVDD**), **PVDD2P** pad for pad ring power, **PVSS1P** pad for (**DVSS, AHVSS, AHVSSG**) and **PVSS2P** pad for pad ring ground. [10]
- If the whole chip is not pad limited, please use more power pads with double (multiple)-bond configuration to reduce power (ground) bouncing due to inductance of bonding wires and fast clock switching.
- Buffers might be used between input I/O pad and PLL Fin input due to constraints on PLL input clock rise/fall time.

Note [10]: **PVDD1P, PVDD1PC, PVDD2P, PVSS1P** and **PVSS2P** pads are power and ground pads of TSMC Analog I/O library. The pad names might be different in different libraries. Please refer to TSMC Analog I/O Application Note.

Placement & Routing

- We suggest placing PG13A1G3 at corner, and keep the power and ground pins of PLL close to the analog power I/O pads. Keep the routing paths as short as possible and far away from any other high drive or simultaneously switching digital I/O pads.
- Don't place noisy or fast output drivers and potentially noisy internal circuits near the PG13A1G3 macro cell. In general, higher frequency signals generate more noise than lower frequency signals. Also, high-power level cells create more noise than low-power level cells.
- Core-logic power rings cannot cross over PG13A1G3. Create dead space between the PG13A1G3 internal circuits and all other internal circuits. Physical space is a good way to reduce noise transmission through the chip substrate. It is recommended that a minimum 30um space between PG13A1G3 and digital core power rings is kept, and 100um space between PG13A1G3 and any other diffusion or well of digital core also.
- Avoid core-logic power lines and signal lines crossing over any power lines of PG13A1G3.
- **Fig 3** is a placement and routing example of PG13A1G3 macro. In this example, **PRDIODE** power cut cells are inserted between power pads of PLL and other digital pads to isolate noise coupling. Double bonding is recommended at the power pad (**AHVDD, AHVSS, AHVDDG, AHVSSG**) to reduce the wire inductance. It is also suggested to minimize the bonding wire length of the power/ground pads if possible.

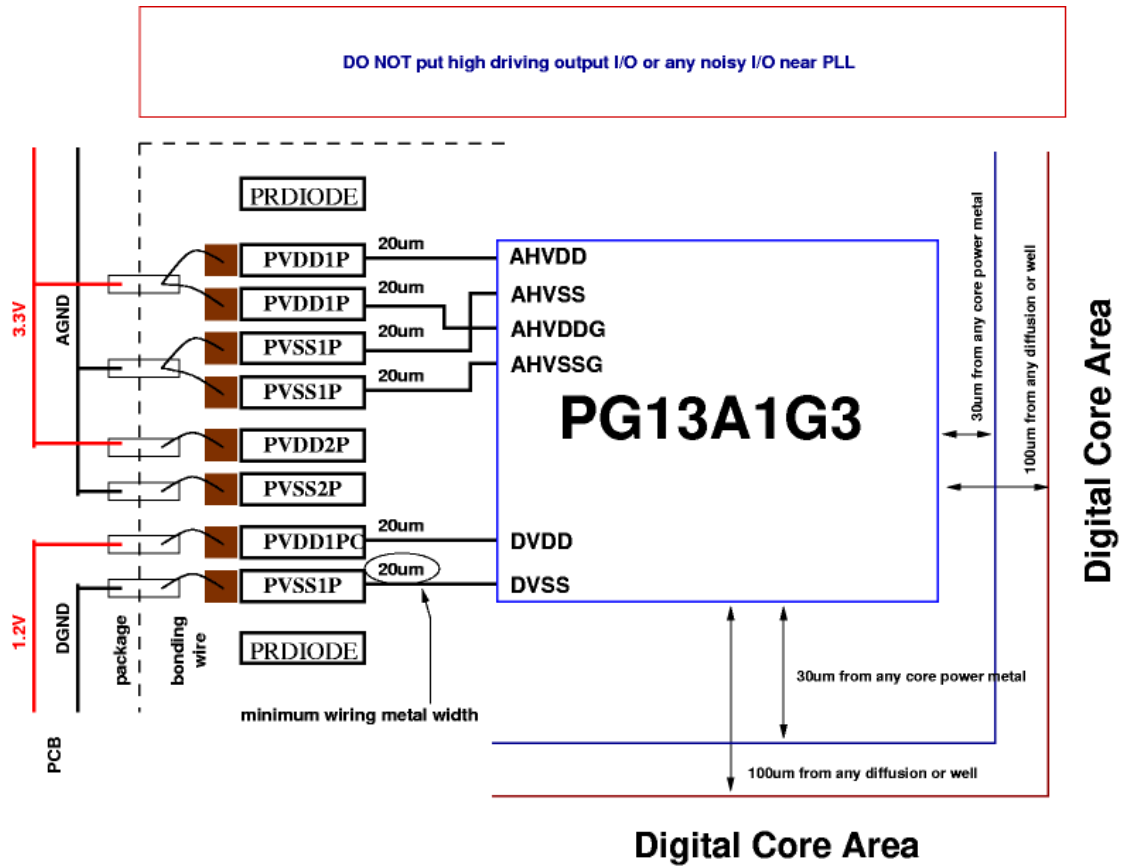


Fig. 3 Placement and Routing Consideration of PLL

Appendix A : PLL Terminology Definition

- **Period Jitter (A), (JEDEC Definition - JESD65):** The edge deviation to the ideal F_{out} when we measure the rising edge of F_{out} after $(n+N)$ -th cycle by using the rising edge of F_{out} at n -th cycle as trigger point, where $N=1$, F_{out} is PLL's output.
- **Short-Term Period Jitter (B):** The edge deviation to the ideal F_{out} when we measure the rising edge of F_{out} after $(n+N)$ -th cycle by using the rising edge of F_{out} at n -th cycle as trigger point, where $N < 10$, F_{out} is PLL's output.
- **Long-Term Period Jitter (C):** The edge deviation to the ideal F_{out} when we measure the rising edge of F_{out} after $(n+N)$ -th cycle by using the rising edge of F_{out} at n -th cycle as trigger point, where N must be large enough (ex, thousands), F_{out} is PLL's output.

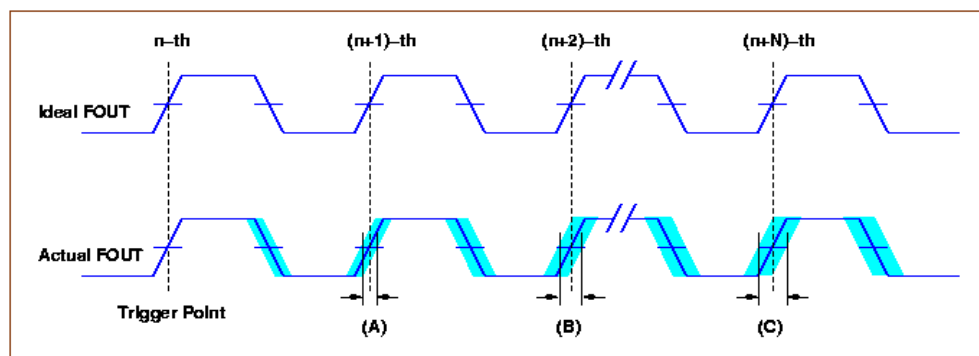


Fig. 4 Definition of Period Jitter

- **Cycle-to-Cycle Jitter (JEDEC Definition - JESD65):** The cycle time variation between adjacent cycles over a random sample of adjacent cycle pairs.

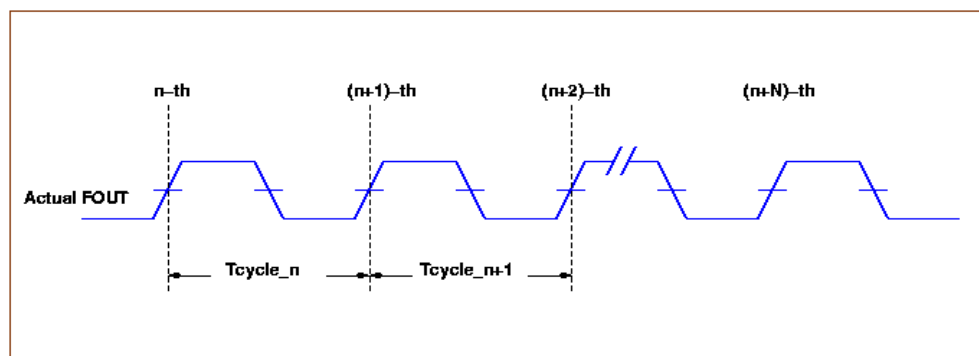


Fig. 5 Definition of Cycle-to-Cycle Jitter

- **Static Phase Error (D):** A fixed skew between the rising edges of 2 synchronized clocks **FIN** and **FBK**.
 - **Static-Phase-Error** > 0 means **FIN** lead **FBK**.
 - **Static-Phase-Error** < 0 means **FBK** lead **FIN**.
- **Tracking Jitter (E), (JEDEC Definition - JESD65):** The edge deviation when we measure the rising edge of **FBK** by using rising edge of **FIN** as trigger point.

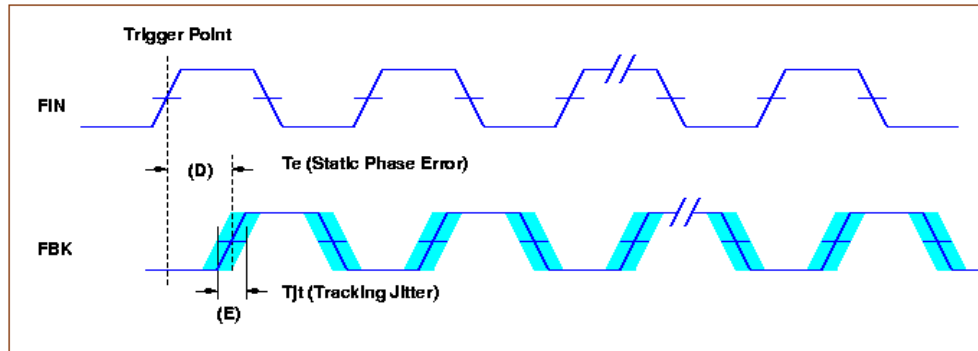


Fig. 6 Definition of Tracking Jitter

- **Peak-to-Peak value and Rms value:**As the **Fig7.** shows, a histogram of FOUT rising edge variation could be resulted from any kind of jitter measurement. The peak-to-peak value is meant by the difference of the max. and min. of the histogram. The rms value is the 1-sigma (standard deviation) value.

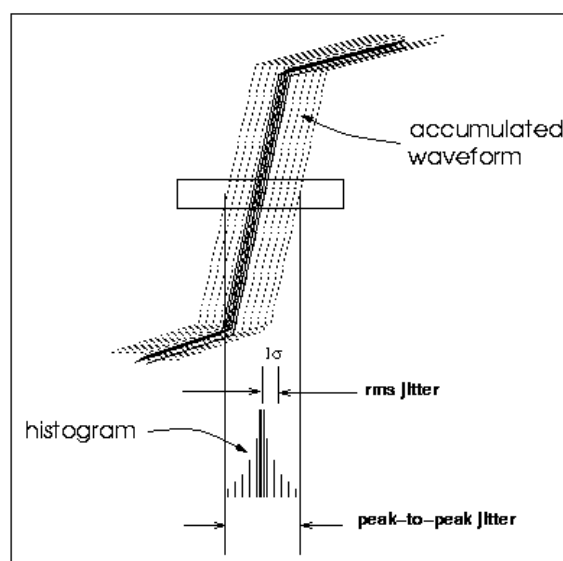


Fig. 7 Jitter Histogram